$$\frac{P}{(\rho g)} + \frac{V^2}{2g} + z = H$$

$$A_1 V_1 = A_2 V_2$$

$$R_e = \frac{\rho v D}{\mu}$$

$$P = P_0 + \rho g h$$

$$h_f = F\left(\frac{L}{D}\right)\left(\frac{V^2}{2g}\right)$$